In vector form, $\vec{\tau} = \left(I\vec{A}\right) \times \vec{B}$

The above equation can also be written in terms of magnetic dipole moment

$\vec{\tau} = \vec{p}_m \times \vec{B}$ where $\vec{p}_m = I\vec{A}$

The tendency of the torque is to rotate the loop so as to align its normal vector with the direction of the magnetic field.

If there are N turns in the rectangular loop, then the torque is given by $\tau = NIAB \sin\theta$

**Special cases:**

(a) When $\theta = 90°$ or the plane of the loop is parallel to the magnetic field, the torque on the current loop is maximum.

$\tau_{max} = IAB$

(b) When $\theta = 0°/180°$ or the plane of the loop is perpendicular to the magnetic field, the torque on the current loop is zero.

## 3.11.2 Moving coil galvanometer

Moving coil galvanometer is a device which is used to detect the flow of current in an electrical circuit.

**Principle** When a current carrying loop is placed in a uniform magnetic field, it experiences a torque.

### Construction

A moving coil galvanometer consists of a rectangular coil PQRS of insulated thin copper wire. The coil contains a large number of turns wound over a light metallic frame. A cylindrical soft-iron core is placed symmetrically inside the coil as shown in Figure 3.59. The rectangular coil is suspended freely between two pole pieces of a horse-shoe magnet.

The upper end of the rectangular coil is attached to one end of fine strip of phosphor bronze and the lower end of the coil is connected to a hair spring which is also made up of phosphor bronze. In a fine suspension strip, a small plane mirror is attached in order

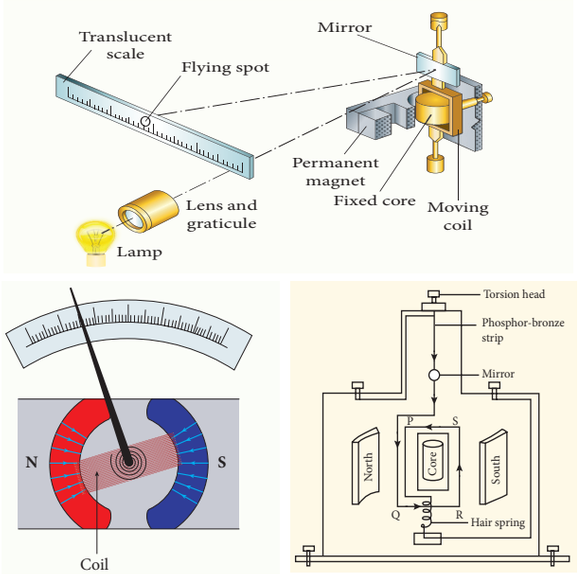



**Figure 3.59** Moving coil galvanometer – its parts

to measure the deflection of the coil with the help of lamp and scale arrangement. The other end of the mirror is connected to a torsion head. In order to pass electric current through the galvanometer, the suspension strip and the spring are connected to terminals.

### Working

Consider a single turn of the rectangular coil PQRS whose length is $l$ and breadth $b$. PQ = RS = $l$ and QR = SP = $b$. Let $I$ be the electric current flowing through the rectangular coil PQRS as shown in Figure 3.60. The horse-shoe magnet has hemi - spherical magnetic poles which produces a radial magnetic field. Due to this radial field, the sides QR and SP are always parallel to the magnetic field $B$ and experience no force. The sides PQ and RS are always perpendicular to the magnetic field and experience equal forces in opposite directions. Due to this, torque is produced.

For single turn, the deflecting torque is

$\tau = bF = bBIl = (lb)\, BI = ABI$

since area of the coil, $A = lb$

For coil with $N$ turns, we get



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